Unit 3 Challenge Exam

Math 9E Parts 1–3 · Lessons 3.1–3.4 20 questions · 98 marks Name: _____

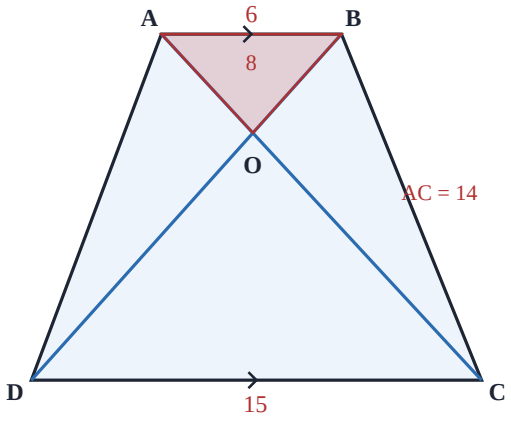
The hardest questions your teacher could ask on this unit: each one combines two or more skills. Show all your work. Answers are on the last pages.

Part 1 · Similar Triangles & Conversions

Question 1

[6 marks]

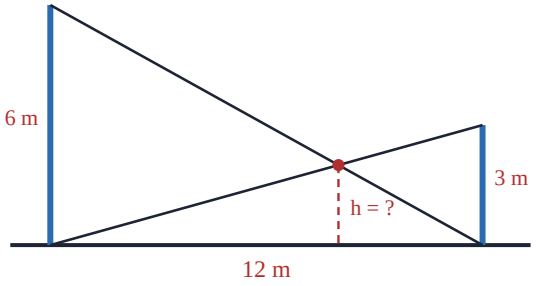
Trapezoid ABCD has AB ∥ DC, AB = 6 cm and DC = 15 cm. The diagonals meet at O, and AC = 14 cm. (a) Prove ΔAOB ~ ΔCOD. (b) Find AO and OC. (c) The area of ΔAOB is 8 cm². Find the areas of ΔCOD and ΔAOD.



Question 2

[5 marks]

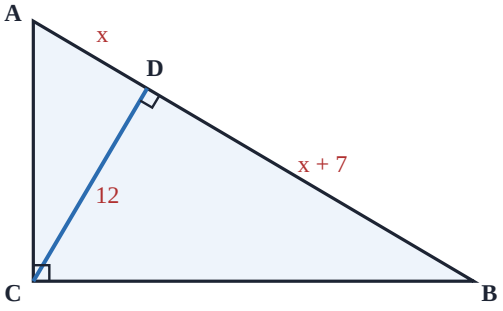
Two vertical poles 6 m and 3 m tall stand 12 m apart. A wire runs from the top of each pole to the bottom of the other. (a) How high above the ground do the wires cross? (b) How far from the taller pole is the crossing point? (c) Show that this height does not depend on the distance between the poles.



Question 3

[5 marks]

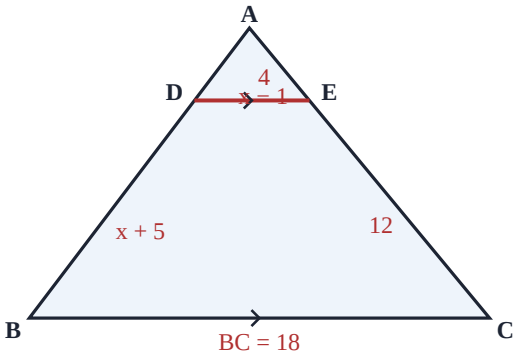
In right ΔABC (right angle at C), CD is the altitude to the hypotenuse AB. AD = x, DB = x + 7 and CD = 12. Find x, then AC, BC and the area of ΔABC.



Question 4

[5 marks]

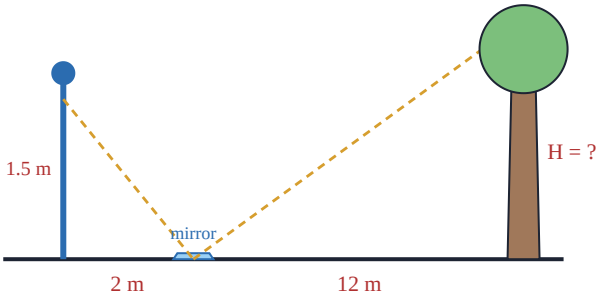
In ΔABC, DE ∥ BC with D on AB and E on AC. AD = x − 1, DB = x + 5, AE = 4, EC = 12 and BC = 18. (a) Find x. (b) Find DE. (c) Find the ratio of the area of ΔADE to the area of trapezoid DECB.



Question 5

[4 marks]

A student's eyes are 1.5 m above the ground. She places a mirror flat on the ground 2 m from her feet and walks until she sees the top of a tree in it. The mirror is 12 m from the base of the tree. (a) Explain why the two triangles are similar. (b) Find the height of the tree.



Question 6

[4 marks]

A doctor orders Gentamicin at 4 mg per kg of body mass per day for a 154 lb patient, given in equal doses every 8 hours (1 kg = 2.2 lb). The vial label reads 80 mg in 2 mL. How many mL should each dose be, to the nearest hundredth?

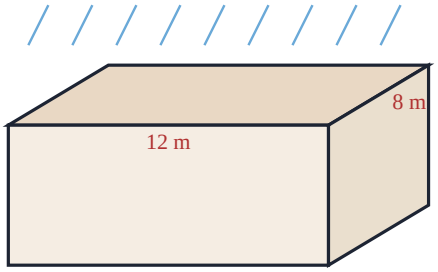


Patient: 154 lb
4 mg/kg/day
every 8 h

Question 7

[4 marks]

During a storm, 2 cm of rain falls on a flat roof measuring 12 m by 8 m. All of it drains into a tank. How many litres reach the tank? How many US gallons is that (1 gal = 3.785 L)?



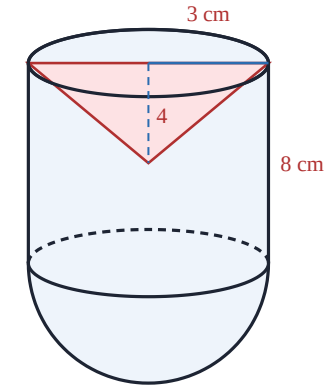
2 cm of rain on a flat roof

Part 2 · Surface Area & Volume

Question 8

[6 marks]

A solid cylinder (radius 3 cm, height 8 cm) has a cone of height 4 cm carved out of its top; the cone's base is the whole top face. A solid hemisphere is attached to the bottom. Find (a) the volume and (b) the total surface area, in terms of π and to the nearest tenth.

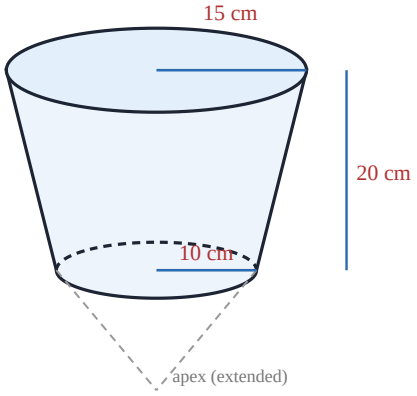


cone carved out · hemisphere added

Question 9

[6 marks]

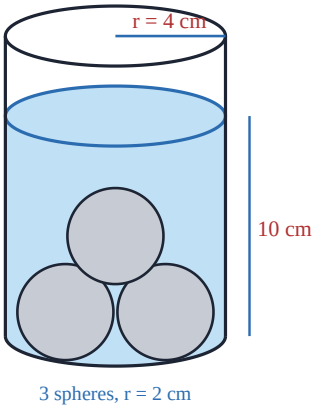
A bucket is a cone with its tip cut off: top radius 15 cm, bottom radius 10 cm, depth 20 cm. Use similar triangles to find the height of the full cone, then find the bucket's capacity in litres.



Question 10

[4 marks]

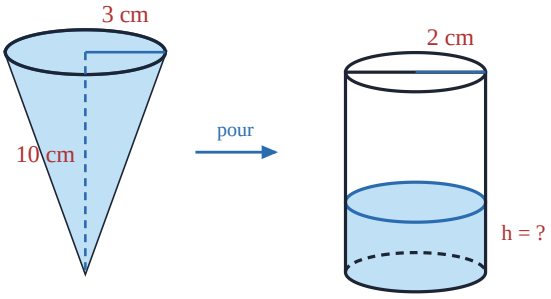
A cylindrical glass of radius 4 cm holds water 10 cm deep. Three metal spheres of radius 2 cm are dropped in and sink completely. How much does the water level rise, and what is the new depth?



Question 11

[4 marks]

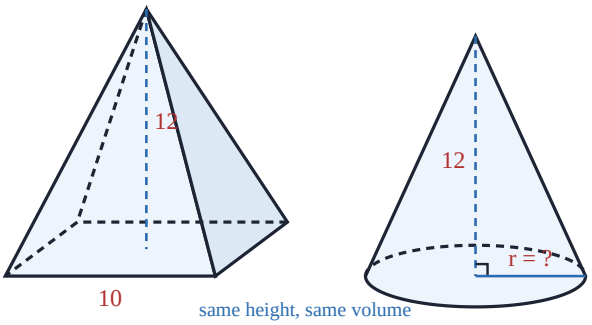
A conical cup (radius 3 cm, height 10 cm) is full of water. The water is poured into an empty cylinder of radius 2 cm. How deep is the water in the cylinder?



Question 12

[6 marks]

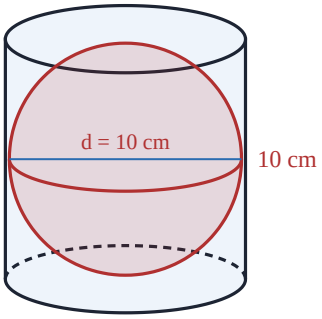
A square pyramid (base side 10 m) and a cone have the same height, 12 m, and the same volume. (a) Find the radius of the cone. (b) Which solid has the larger lateral (side) surface area, and by how much?



Question 13

[5 marks]

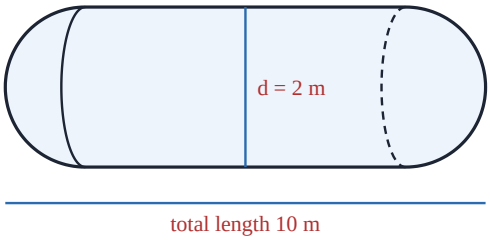
A ball of diameter 10 cm fits exactly inside a cylindrical can: it touches the top, the bottom and the sides. (a) What percentage of the can is empty space? (b) Compare the ball's surface area with the can's curved (side) surface area.



Question 14

[5 marks]

A fuel tank is a cylinder with a hemisphere on each end. Its total length is 10 m and its diameter is 2 m. Find (a) its outer surface area and (b) its capacity in kL.

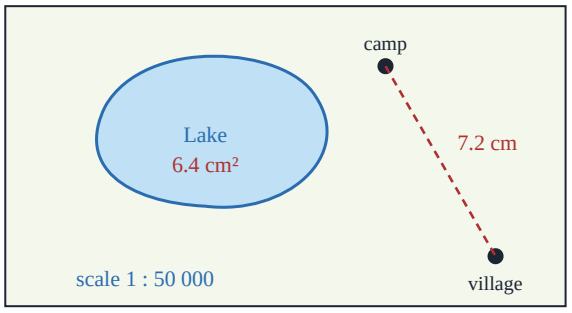


Part 3 · Scale & Scale Diagrams

Question 15

[5 marks]

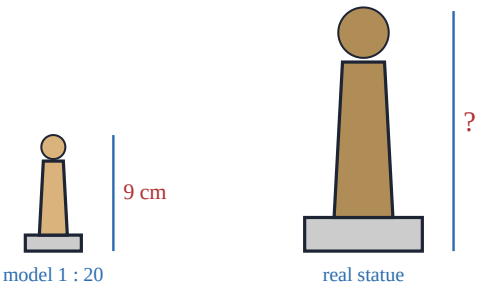
On a map with scale 1 : 50 000, a lake covers 6.4 cm² and the trail from the camp to the village measures 7.2 cm. (a) Find the real length of the trail in km. (b) How long does it take to walk at 4.8 km/h? (c) Find the real area of the lake in km² and in hectares.



Question 16

[5 marks]

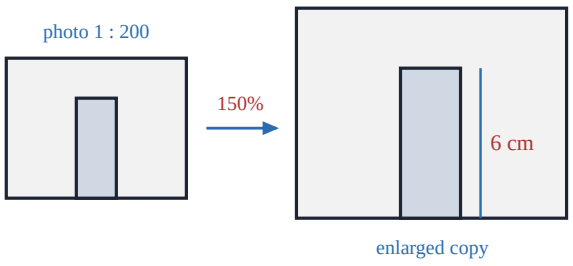
A clay model of a statue is built at a scale of 1 : 20. The model is 9 cm tall, has a surface area of 240 cm^2 and uses 150 cm^3 of clay. Find the real statue's (a) height in m, (b) surface area in m^2 and (c) volume in m^3.



Question 17

[4 marks]

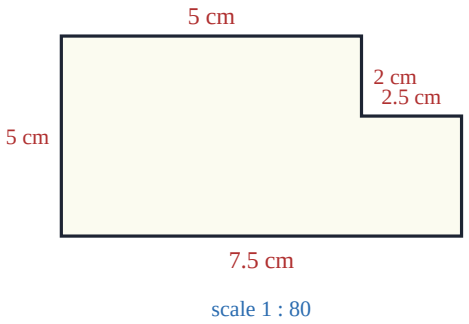
A photo of a building has a scale of 1 : 200. It is enlarged to 150% on a copier. (a) What is the scale of the enlarged copy? (b) The building is 6 cm tall on the copy. How tall is the real building?



Question 18

[5 marks]

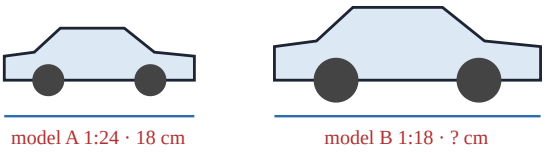
An L-shaped room is drawn at a scale of 1 : 80. On the plan, the room is a 7.5 cm × 5 cm rectangle with a 2.5 cm × 2 cm corner cut out. (a) Find the real floor area in m^2. (b) Carpet costs \$28/m^2 plus 13% tax. Find the total cost.



Question 19

[5 marks]

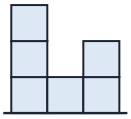
Model A of a car is built at 1 : 24 and is 18 cm long. Model B of the same car is built at 1 : 18. (a) How long is the real car in m? (b) How long is Model B? (c) Model A needs 300 cm^2 of paint. How much does Model B need?



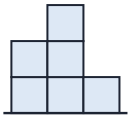
Question 20

[5 marks]

A solid is built from 1-cm cubes. Its front view shows columns of 3, 1 and 2 cubes (left to right) and its right view shows 2, 3 and 1 cubes (front to back). The top view is unknown. (a) What is the greatest number of cubes the solid could have? (b) What is the least number? Explain.

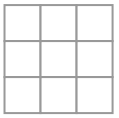


front view



right view

top view ?



ANSWER KEY

Part 1 · Similar Triangles & Conversions

Q1 (a) AB ∥ DC ⇒ ∠OAB = ∠OCD and ∠OBA = ∠ODC (alternate angles) ⇒ ΔAOB ~ ΔCOD (AA). (b) $k = 15/6 = 2.5$, so $OC = 2.5 \cdot AO$. $AO + 2.5 \cdot AO = 14$ ⇒ **AO = 4 cm, OC = 10 cm**. (c) Area COD = 8×2.5^2 = **50 cm²**. ΔAOD and ΔAOB have the same height from A to line BD, and $OD = 2.5 \cdot OB$ ⇒ area AOD = 8×2.5 = **20 cm²**.

Q2 Let the crossing be h high and x from the taller pole. Wire from the short pole's top to the tall pole's foot: $h/x = 3/12$. Wire from the tall pole's top to the short pole's foot: $h/(12 - x) = 6/12$. So $x = 4h$ and $12 - x = 2h$ ⇒ $6h = 12$ ⇒ **h = 2 m**, **x = 8 m**. (c) For poles a and b, distance d: $x = hd/b$ and $d - x = hd/a$ ⇒ $d = hd(1/a + 1/b)$ ⇒ **h = ab/(a + b)**, with no d. Check: $6 \cdot 3/9 = 2$ ✓

Q3 $CD^2 = AD \cdot DB$ ⇒ $x(x + 7) = 144$ ⇒ $x^2 + 7x - 144 = 0$ ⇒ $(x + 16)(x - 9) = 0$ ⇒ **x = 9** (a length is positive). AB = 25. $AC^2 = 9 \cdot 25 = 225$ ⇒ **AC = 15**. $BC^2 = 16 \cdot 25 = 400$ ⇒ **BC = 20**. Area = ½·15·20 = **150**.

Q4 (a) $(x - 1)/(x + 5) = 4/12$ ⇒ $12x - 12 = 4x + 20$ ⇒ **x = 4** (AD = 3, DB = 9, AB = 12). (b) DE/BC = AD/AB = 3/12 ⇒ DE = 18/4 = **4.5**. (c) $k = 1/4$ ⇒ area ADE = $(1/4)^2 = 1/16$ of ΔABC, so the trapezoid is 15/16 ⇒ ratio **1 : 15**.

Q5 (a) Light reflects at equal angles, so the angle at the mirror is the same in both triangles, and the person and the tree both stand at 90° to the ground ⇒ similar (AA). (b) $1.5/2 = H/12$ ⇒ **H = 9 m**.

Q6 154 ÷ 2.2 = 70 kg. Daily dose = 4×70 = 280 mg. Every 8 h = 3 doses ⇒ 93.33 mg per dose. Volume = $93.33 \times 2/80 \approx$ **2.33 mL**.

Q7 $V = 12 \times 8 \times 0.02 = 1.92$ m³ = **1920 L** (1 m³ = 1000 L). 1920 ÷ 3.785 ≈ **507.3 gal**. Trap: convert 2 cm to 0.02 m first.

Part 2 · Surface Area & Volume

Q8 (a) $V = \pi(9)(8) - \frac{1}{3}\pi(9)(4) + \frac{2}{3}\pi(27) = 72\pi - 12\pi + 18\pi$ = **78π ≈ 245.0 cm³**. (b) Cone slant = $\sqrt{(3^2 + 4^2)} = 5$. SA = side $2\pi(3)(8)$ + cone $\pi(3)(5)$ + hemisphere $2\pi(9) = 48\pi + 15\pi + 18\pi$ = **81π ≈ 254.5 cm²** (no flat top is left).

Q9 $H/(H - 20) = 15/10$ ⇒ $15H - 300 = 10H$ ⇒ **H = 60 cm** (the removed tip is 40 cm tall). $V = \frac{1}{3}\pi(15^2)(60) - \frac{1}{3}\pi(10^2)(40) = 4500\pi - 1333.3\pi \approx 3166.7\pi \approx 9948$ cm³ ≈ **9.95 L**.

Q10 Volume of spheres = $3 \times (4/3)\pi(2^3) = 32\pi$ cm^3. Rise = $32\pi \div (\pi \cdot 4^2)$ = **2 cm**. New depth = **12 cm**.

Q11 $V_{cone} = \frac{1}{3}\pi(3^2)(10) = 30\pi$ cm^3. In the cylinder: $\pi(2^2)h = 30\pi \Rightarrow$ **h = 7.5 cm**.

Q12 (a) $\frac{1}{3}(10^2)(12) = \frac{1}{3}\pi r^2(12) \Rightarrow r^2 = 100/\pi \Rightarrow$ **r ≈ 5.64 m**. (b) Pyramid slant = $\sqrt{(12^2 + 5^2)} = 13 \Rightarrow$ lateral = $4 \cdot \frac{1}{2} \cdot 10 \cdot 13 = 260$ m^2. Cone slant = $\sqrt{(12^2 + 31.83)} \approx 13.26 \Rightarrow$ lateral = $\pi(5.64)(13.26) \approx 235.0$ m^2. The **pyramid** is larger by about **25.0 m²**.

Q13 r = 5, h = 10. $V_{can} = \pi(25)(10) = 250\pi$; $V_{ball} = (4/3)\pi(125) \approx 166.7\pi = \frac{2}{3}$ of the can ⇒ **33.3% empty**. (b) Ball: $4\pi(5^2) = 100\pi$. Can side: $2\pi(5)(10) = 100\pi \Rightarrow$ **they are equal** (Archimedes' result).

Q14 r = 1 m; cylinder part = 10 − 2 = 8 m. (a) SA = $2\pi(1)(8) + 4\pi(1^2)$ = **20π ≈ 62.8 m²** (two hemispheres = one sphere). (b) V = $\pi(1^2)(8) + (4/3)\pi(1^3) = (28/3)\pi \approx 29.3$ m^3 = **29.3 kL**.

Part 3 · Scale & Scale Diagrams

Q15 (a) 7.2 × 50 000 = 360 000 cm = **3.6 km**. (b) 3.6 ÷ 4.8 = 0.75 h = **45 min**. (c) Area factor $50\,000^2 = 2.5 \times 10^9$: $6.4 \times 2.5 \times 10^9 = 1.6 \times 10^{10}$ cm^2 = 1.6×10^6 m^2 = **1.6 km² = 160 ha**.

Q16 (a) 9 × 20 = 180 cm = **1.8 m**. (b) $240 \times 20^2 = 96\,000$ cm^2 = **9.6 m²**. (c) $150 \times 20^3 = 1\,200\,000$ cm^3 = **1.2 m³**.

Q17 (a) 1.5 : 200 = **3 : 400** (about 1 : 133.3). (b) 6 × 400/3 = 800 cm = **8 m**. Check: on the original photo it is 6 ÷ 1.5 = 4 cm, and 4 × 200 = 800 cm ✓

Q18 Real sizes: 7.5 × 80 = 600 cm = 6 m; 5 × 80 = 4 m; cut-out 2 m × 1.6 m. (a) Area = 6 × 4 − 2 × 1.6 = **20.8 m²**. (b) 20.8 × 28 = \$582.40; × 1.13 = **\$658.11**.

Q19 (a) 18 × 24 = 432 cm = **4.32 m**. (b) 432 ÷ 18 = **24 cm**. (c) Scale factor A → B = 24/18 = 4/3 ⇒ area × $(4/3)^2 = 16/9 \Rightarrow 300 \times 16/9 \approx$ **533.3 cm²**.

Q20 Use a 3 × 3 top grid: columns take the front heights (3, 1, 2), rows take the right-view heights (2, 3, 1). (a) Each square can hold at most min(column, row): row 2: 2 + 1 + 2 = 5; row 3: 3 + 1 + 2 = 6; row 1: 1 + 1 + 1 = 3 ⇒ **14 cubes**. (b) Each view height needs only one stack: put 3 where the 3-row meets the 3-column, 2 where the 2-row meets the 2-column, 1 where the 1-row meets the 1-column ⇒ 3 + 2 + 1 = **6 cubes**.